GG-S-945a
DECEMBER 4, 1959
SUPERSEDING
Int. Fed Spec. GG-S-00945 (VA-MI)
August 26, 1958

FEDERAL SPECIFICATION

SYRINGE AND NEEDLE, DISPOSABLE, HYPODERMIC, STERILE, SINGLE INJECTION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal agencies

1 SCOPE AND CLASSIFICATION

1 1 **Scope** - This specification covers sterile disposable syringes, which are suitable for a single injection of parenteral drugs or solutions or for obtaining single specimens

1.2 Classification

1.2.1 *Types and styles* - Sterile disposable syringes, covered by this specification, shall be of one type and grade and of the following styles, as specified in the procurement documents

Style I - Syringe without needle
Style II - Syringe with needle affixed

1.2.2 *Sizes*

1.2.2 1 *Syringe sizes* - Sterile disposable syringes, covered by this specification, shall be of the following sizes, as specified

Size 1 - 2-cubic-centimeter syringe
Size 2 - 5-cubic-centimeter syringe
Size 3 - 10-cubic-centimeter syringe

1 2.2 2 *Needle sizes* - Needles for style II syringes shall be of the following sizes, as specified

Size A - 20-gage, 1-1/2-inch needle
Size B - 22-gage, 1-1/2-inch needle
Size C - 25-gage, 5/8-inch needle

2 APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2 1 **Specifications and standards** - The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification

Federal Specifications

PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-676 - Boxes, Set-Up, Paperboard

FSC 6515

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Federal Standards

Fed Std No 102 - Preservation, Packaging and Packing Levels
Fed Std No 123 - Marking for Domestic Shipment (Civilian Agencies)

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S Government Printing Office, Washington 25, D C

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo , Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D C

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies)

Military Specifications

MIL-B-4229 - Boxes, Paperboard, Metal-Stayed
MIL-L-10547 - Liners, Case, Waterproof

Military Standards

MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

2.2 Other publications - The following documents form a part of this specification Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply

American Standards Association, Inc Publications

Z70 1 - 1955 (UDC615 47) - American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applictions

(Application for copies should be addressed to the American Standards Association, Incorporated, 70 East Forty-Fifth Street, New York 17, New York)

United States Pharmacopoeial Convention, Inc

Pharmacopoeia of the United States

(Application for copies should be addressed to the Mack Printing Company, Easton, Pennsylvania)

3. REQUIREMENTS

3 1 Material

3 1.1 *Plastic* - Plastic shall be virgin, nontoxic, high grade throughout It shall be free from

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reclaimed or reground material and shall not have more than 10 p p m extractable heavy metals (see 4 3 6) Plastic shall be polystyrene, high-impact polystyrene, polyethylene, or other plastic suitable in all respects for the purpose intended

3 1.2 *Glass* - Glass shall be high grade, corrosion-resistant throughout

3 1.3 *Steel* - Steel shall be corrosion-resisting steel conforming to the following chemical composition

Chromium, 16 to 20%	Silicon, 0 50% (maximum)
Nickel, 7 to 10%	Sulphur, 0 025% (maximum)
Carbon, 0 08% (maximum)	Phosphorus, 0 025% (maximum)
Manganese, 0 50% maximum	

As an alternate to the above, the material shall be in conformance with type 302, AISI stainless steel

3 1 4 *Rubber* - Rubber shall be fresh stock, high pharmaceutical grade, nonblooming, nonleaching, nontoxic, age resistant natural rubber or neoprene or combinations thereof processed for the specific purpose (see 4 3 6)

3.1.5 *Lubricant and coating of barrel and plunger gasket* - Shall be an inert, nontoxic, silicone compound suitable for the specific use It shall also serve as a siliconizing coating on all interior surfaces of the syringe that may come in contact with the liquid content, while being used for making an injection Silicone compound shall be of a high pharmaceutical grade and nontoxic to any portion of blood samples which may be taken in or stored in syringes during a normal hospital laboratory procedure (see 4 3 6)

3 1.6 *Cotton* - When cotton is used as protective filter on open-tip type needle protector, cotton shall be strict middling grade, bleached cotton of staple length, 1-1/16 inch to 1-1/8 inch (see 3 2 4)

3 2 Design - Style I syringe shall consist of a barrel, plunger, plunger gasket and a syringe tip protector Style II syringe shall consist of a barrel, plunger, plunger gasket, affixed needle, and a needle protector, assembled ready for use

3 2 1 *Barrel* - The barrel shall be made of plastic or glass (see 3 1 1 and 3 1 2) Barrel shall be a tube, closed at one end except for an orifice, open at the other end Barrel shall provide good visibility of contents and shall be within the dimensions specified in 3 3 1 and in table I At the closed end there shall be formed a tip for the attachment of a syringe needle The wall at the open end shall be thickened and extended radially outward to form a flange to prevent slipping of the barrel through the fingers The inside surface of the barrel shall be a very fine uniform surface, free from pits, grind marks, or open air lines Inside surface of the barrel shall be coated with a silicone compound as specified in 3 1 5

3.2.1 1 *Flange* - The form and dimensions of the flange shall be appropriate to the size of the barrel, in all cases shall extend outward from the wall sufficiently to give ample fingerhold and lengthwise of the barrel to give strength and durability Opposite segments of the flange shall be removed to provide straight parallel sides of ample length to prevent rolling

3.2 1.2 *Tip* - Unless otherwise specified in the invitation for bids, the tips shall be formed concentric to the axis of the barrel Orifice through tip shall be concentric therewith and shall be within the specifications of the dimensions of table I The junction of the syringe needle to the tip shall be fluid tight when tested in accordance with 4 3 1 and 4 3 1 1 The Luer slip tips shall fit the ring gages as shown in figure 7 of ASA Standard Z70 1-1955

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3.2.1.3 *Graduations and other markings* - The outside surface of the barrel shall have marked thereon the graduation scale required for this type syringe, the words "Discard After Use", or other similar legend denoting one time use and such other markings as the invitation for bids may require Main graduation marks shall be at least 33-1/3 percent longer than the subgraduation marks, and the "0" mark of the scale shall register precisely with the extreme distal end of the plunger gasket when pushed to the bottom of the barrel The ungraduated portion shall be not greater than one-third of the overall length of the syringe Marks shall be of red, or other dark color, legible material that will not rub off or be marred during use or shipment The marking pigment or ink shall withstand the tests required under 4 3 3

3.2.2 *Plunger* - Shall be either hollow or solid The distal end of the plunger shall be shaped to permit attachment of a plunger gasket The proximal end of the plunger shall extent well out of the barrel and shall be shaped to form a knob of a thickness and diameter appropriate to the size of the syringe or shaped to allow a suitable finger grip for usage and of a size appropriate to the size of the syringe The distal end of each plunger with its attached plunger gasket shall accurately fit its individual barrel so as to make the syringe successfully stand the tests prescribed in 4 3 1, 4 3 1 1 and 4 3 1 2, and yet permit the plunger and plunger gasket to draw and move freely and smoothly throughout its excursion of the barrel

3.2.2 1 *Plunger gasket* - Shall be manufactured of material specified in 3 1 4 and designed and processed for the purpose to fit the barrel so accurately as to make the syringe successfully stand the tests prescribed in 4 3 1, 4 3 1 1 and 4 3 1 2 Plunger gasket shall not become loose from plunger after being moistened with medication Shall be properly treated with a silicone lubricant An excess of lubricant that detracts from appearance, or can be caused to form pools or droplets, or can be washed out with the solution being used, is cause for rejection (See 4 3 7) The distal end of the gasket shall be designed or beveled sufficiently to permit easy and efficient removal of air bubbles from fluid path In any case, graduation markings denoting correct, accurate capacity shall register precisely with the first ring at the plunger gasket distal end, touching sides of the barrel Shall be of a permanently suitable color so that the first ring at the plunger gasket distal end, touching sides of the barrel, shall be clearly visible and shall accurately indicate through the barrel wall

3 2.3 *Needle* - Shall consist of cannula and hub

3 2.3 1 *Cannula* - Cannula shall be seamless drawn tubing of corrosion-resistant steel in accordance with paragraph 3 1 3 Shall be within the dimensions specified in 3 3 2 1, 3 3 2 2, 3 3 2 3, and table II Shall be securely swaged into either the hub or barrel tip and withstand tests prescribed by 4 3 1, 4 3 1 1, 4 3 2, 4 3 2 1, 4 3 2 2, 4 3 2 3, 4 3 2 4 and 4 3 7

3.2.3 1.1 *Point* - Shall be regular bevel (for subcutaneous and intramuscular injection) and shall be of the length specified in table II for the gage needle specified in invitation for bids The bevel may be ground concave or flat and may have a center ridge at the distal end to strengthen the tip, shall be so ground with a side bevel at the outer end as to enhance the keenness of the tip and lateral cutting edges The outer edges of the ground surface shall be keen and both outer and inner edges shall be free from burrs and slivers The tip shall be straight and of maximum keenness

3.2.3.2 *Hub* - If hub is used, it shall be aluminum, plastic, or plated-brass hub Brass hub shall be chromium or nickel plated Shall be formed and shaped to fit precisely the barrel tip to which it is to be securely fastened to prevent dislodgement and withstand the test in compliance with 4 3 1, 4 3 1 1, 4 3.2, and 4 3 2 6

3.2 4 *Protector, needle and syringe tip* - Shall be of plastic or rubber so designed as to fit securely over tip Shall be so designed that it shall not fall off or work loose from the syringe under ordinary shipping conditions or handling When end of tip is open, it shall contain, at its distal end, a suitably

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supported filter, made of material specified in 3 1 6, which will protect the fluid path of the needle and syringe and maintain its sterility during shipment and ordinary handling until time of use Shall be so designed as to prevent it from shattering when used as a means of breaking off tip to render syringe unusable

3.3 Dimensions.

3.3 1 Dimensions of the syringe without the needle cannula and hub shall conform to table I

TABLE I - Barrel

Capacity, plus or minus 5 percent	Length overall, including tip	Diameter outside	Diameter inside	Diameter orifice	Subgraduations	Main graduation
cc	Inches	Inches	Inches	Inches	cc	cc
2	2 1 - 2 9	0 4 - 0 5	0 3 - 0 4	0 015 - 0 030	0 1	0 5
5	2 75 - 3 5	5 - 75	45 - 0 6	015 - 080	2	1 0
10	3 75 - 4 3	6 - 85	55 - 0 7	015 - 080	2	1 0

3.3 1 1 Graduation scale shall be in cubic centimeters or decimals thereof unless otherwise specified in the invitation to bid

3.3 2 Dimensions of needle cannula shall conform to table II

TABLE II - Dimension of needles

Gage (1)	Point	Cannula			
	Length	Length	Outside diameter		Inside diameter (Lumen) measured at point, minimum (6)
	Tolerance, 10 percent (2)	(3)	Maximum (4)	Minimum (5)	
	Inches	*Inches*	*Inches*	*Inches*	*Inches*
25	0 094	5/8 ± 3/64	0 0205	0 0195	0 0095
22	133	1-1/2 ± 1/16	0285	0275	0155
20	164	1-1/2 ± 1/16	036	034	022

3.3.2.1 *Length* - The length of the cannula (measured from the tip of point to the junction with the hub or barrel tip) shall be as specified in table II, column 3 The cannula shall have additional length to extend well into the hub or barrel tip The length shown in table II, column 3, shall be specified as the length of the needle in the invitation for bids

3.3.2.2 *Diameter* - Outside diameter of the needle called for in the invitation for bids, shall be specified for the appropriate stock number by gage number, and shall be as in table II, columns 4 and 5 Inside diameter shall be as in column 6

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3.3.2.3 *Length of point* - (Measured from heel to tip paralleled with axis of cannula) Shall be as in table II, column 2

3 4 Sterility

3 4 1 *Sterility procedure* - Each syringe, as specified, shall be sterilized by any commercially acceptable procedure

3 4.2 *Certificate of sterility* - A certificate of sterility, listing all lot or control numbers stating that samples from each sterilizer charge have been tested and found sterile, shall be submitted by the contractor to the procuring activity

3 4.2 1 *Test for sterility* - At the option of the procuring authority, and prior to final acceptance, samples taken in 4 1 shall be tested for sterility as set forth in 4 3 4

3 5 Pyrogens - Contractor shall submit certificate of freedom from pyrogens, covering each lot with lot or control numbers indicated

3.5 1 *Test for freedom from pyrogens* - At the option of the procuring authority, and prior to final acceptance, samples taken in paragraph 4 1 shall be tested for freedom from pyrogens as set forth in 4 3 5

3 6 Workmanship - Shall be first-class in every respect, free from any defect affecting appearance, safety, or use for which it is intended The fluid path of the syringe and needle shall be free from lint, dirt, or other particulate matter, when inspected and tested as described in 4 2 and 4 3 7

4 SAMPLING, INSPECTION AND TEST PROCEDURES

4 1 Sampling - One-half percent of 1 percent of delivery, but not less than five samples, shall be taken at random from each 1,000 syringes of the same size or fraction thereof, except when the shipment is over 10,000 syringes of the same size, in which case at least two samples shall be taken from each one-tenth of the shipment, for examination for compliance with this specification

4.2 Inspection - Samples shall be inspected for strict compliance with the requirements of this specification

4.3 Tests

4.3.1 *Pressure tests* - To conduct the following leakage tests, a pressure of 50 pounds per square inch for the 2-cc size syringe or a pressure of 60 pounds per square inch for the 5-cc and 10-cc syringe shall be used A hypodermic needle of appropriate size shall be attached to the style I syringe when performing the tests

4.3 1.1 *Leakage* - Pressure used shall be as specified in 4 3 1 The water used as a testing fluid shall be let in through the needle until the pressure is raised to the value specified in 4 3 1 The plunger shall be drawn to a point 1/2 the graduated length where it shall be allowed to remain for 10 seconds The plunger shall then be drawn back to midway between first position and the top graduation mark where it shall be allowed to remain for a similar length of time The plunger shall then be drawn back to the top graduation mark where it shall be allowed to remain for a similar length of time The syringe shall be tested in a horizontal position During each test, enough water shall not pass the connection of the cannula with the hub, the connection of the hub with the syringe tip, or connection of the cannula with syringe tip where no hub is used, or plunger gasket to produce a falling drop After each successive test, any water that may have passed shall be wiped away and the next test applied

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4.3.1.2 *Suction test* - The water used as a testing fluid shall be drawn in through the needle in the amounts specified in table III for each size syringe The needle end shall then be sealed and the plunger withdrawn to the top graduation mark There shall be no leakage of air past the plunger gasket

TABLE III - *Suction test*

Syringe Size	Volume of Water
cc	cc
2	0 5
5	2 0
10	5 0

4.3.2 *Needle test, style II syringe*

4.3 2 1 *Security of attachment of cannula to syringe* - The barrel of the syringe shall be firmly held in a suitable fixture with the tip and needle free and exposed A tensile load indicated in table IV required for the different gage needle shall be applied to the needle and allowed to remain for one minute without pulling out of the hub and/or syringe

TABLE IV

Gage	Pull to be applied
	Pounds
25	5
22	5
20	10

4 3 2 2 *Stiffness* - Shall be such that the cannula lying across test edges, separated distances indicated in table V, column 4, and the subjected to bending loads applied in the middle of the span as in column 2 shall bend not to exceed the distance indicated in column 3

TABLE V

Gage	Stiffness			Elasticity		Breakage
	Under load of	Shall bend maximum	Separation of test edges	Without permanent set shall bend	Contact distance, from hub	Without breaking, shall bend 20 times both sides of vertical
(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Ounces	*Inch*	*Inch*	*Degrees*	*Inch*	*Degrees*
25	25	0.026	1/2	12	7/16	25
22	50	029	5/8	12	11/16	25
20	50	021	3/4	13	1	25

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4 3.2 3 *Elasticity* - Shall be such that with the hub or barrel tip in fixed position, the cannula, after bending to one side as in table V, columns 5 and 6, and being released, shall show no permanent set,i e , shall return fully to starting point

4.3.2 4 *Resistance to breakage* - Shall be such that with the hub or barrel tip fixed the cannula shall not break when bent as indicated in table V, column 7

4.3.2.5 *Corrosion resistance* - Cannula shall withstand immersion in 10 percent citric acid at room temperature for 5 hours and, after removal from the acid, boiling for 30 minutes in distilled water, followed by immersion in distilled water at room temperature for 48 hours, without showing appreciable corrosion after drying by evaporation Glass vessels shall be used

4 3.2.6 *Hub* - Shall stand boiling for 30 minutes and remaining immersed for 24 hours in distilled water without showing appreciable corrosion after drying by evaporation Test to be made in glass vessel

4 3.3 *Test for permanency of pigment* - Syringes shall be soaked in water for three hours without removing pigment Pigment should not show appreciable fading when exposed to direct sunlight for twenty-four hours Pigment should not be easily removed by repeated rubbing with the hand or scratching with the fingernail

4.3 4 *Sterility* - Sterility shall be determined in accordance with the latest revision of U S Pharmacopoeia for solids

4.3.5 *Pyrogen* - The pyrogen test shall be conducted in accordance with the latest revision of U S Pharmacopoeia for pyrogen tests

4.3 6 *Toxicity and heavy metals* - A saline extract of syringes, prepared by auto-claving syringes (minus the tip protector and needle) in normal saline shall be nontoxic in mice when injected intravenously at a level of 50 ml /kg body weight An acid extraction of the intact syringe (minus the tip protector and needle) shall show that the extractable heavy metals such as lead shall be less than 10 p p m

4 3.7 *Cleanliness test*

4 3.7 1 *Test material* - Shall be a high viscosity, soft, smooth-textured white substance or paste Before testing syringe and needle, the test material shall be examined closely under 36 power magnification for foreign coloring or matter Any dark gray or black specks, spots, or streaks shall indicate foreign particles and new test material selected, sampled and examined until suitable A representative sample shall be retained for comparison and control purposes with test specimen

4.3 7.2 *Test* - A quantity of the test material sufficient to perform the test but not less than 1/4 cubic centimeter shall be filled into each of five sample syringes with the plungers withdrawn Plunger is carefully inserted and test material forced slowly through each syringe Test specimens shall be first deposit forced through each syringe and spread on white bond paper Presence of foreign matter in two or more syringes, determined by comparison with test material sample, both under 7-power magnification and under physical observation without magnification, using ordinary laboratory lighting, shall be cause for rejection

5 PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing protection used are defined in Federal Standard No 102

5 1 For civil agencies

5 1.1 *Packaging*

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5.1 1 1 *Unit of issue,*- One box or carton of one hundred (100) syringes of the type and size specified constitutes one unit

5 1 1.2 *Packaging* - Each syringe shall be individually and securely packaged in a suitable container to maintain sterility of the syringe until container is opened One hundred packaged syringes (unit of issue) shall be packed in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the syringe for its intended purpose

5.1.2 *Packing* - The number of unit container(s) constituting the quantity ordered for one destination shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose to insure acceptance by common or other carrier for safe transportation without assessment of penalty charges for improper packing

5 1.3 *Marking* - Each syringe container and unit of issue container shall be marked or labeled with the applicable commercial description, (including capacity of syringe, gage and length of needle), lot or control number, contractor's name and the word "Sterile", using contrasting colors and in such a clear manner as to be easily read

5 1 3.1 *Shipping containers* - Shipping containers shall be marked in accordance with Federal Standard No 123

5 2 For military agencies

5.2.1 *Packaging*

5.2 1.1 *Unit of issue* - One hundred syringes as specified constitutes one unit of issue

5 2 1.2 *Packaging quantities* - The number of units of issue indicated in table VI shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified in procurement document

TABLE VI

Style	Size	Unit Package	Inter-mediate Package	Exterior Con-tainer
I	2cc	1 unit	5 units	10 units
I	5 cc	1 unit	5 units	10 units
I	10 cc	1 unit	5 units	10 units
II	2 cc	1 unit	not required	10 units
II	5 cc	1 unit	not required	10 units
II	10 cc	1 unit	not required	10 units

5.2 1.2 1 *Packing variation permitted* - If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination

5.2.1.3 *Level A*

5.2.1.3 1 *Unit packaging* - Each syringe shall be individually and securely packaged in a suitable container to maintain sterility of the syringe until container is opened Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP-B-676 or Military Specification MIL-B-4229 Carton shall contain suitable die-cut inserts, or sufficient

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cushioning material to insure against shifting and breakage Each carton shall be adequately secured to prevent accidental opening

5 2 1 3.2 *Intermediate packaging* - The number of units specified in table VI shall be packaged in a carton of appropriate size, constructed in accordance with Federal Specification PPP-B-566, PPP-B-676 or Military Specification MIL-B-4229 Each carton shall be adequately secured to prevent accidental opening

5 2 1 4 *Level C* - Each syringe shall be individually and securely packaged in a suitable container to maintain sterility of the syringe until container is opened Each unit shall be packaged in standard commercial containers of the size and kind commonly used for the purpose, which will afford the degree of protection required for shipment and use of the end product for its intended purpose within the continental limits of the United States of America

5.2 2 *Packing*

5.2 2.1 *Level A*

5 2.2 1 1 *Exterior container* - The number of units specified in table VI shall be overpacked in an exterior container constructed in accordance with Federal Specification PPP-B-621, table IV

5 2.2 1.2 *Case liner* - Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547 Closure and sealing shall conform to applicable paragraphs of appendix thereto

5.2.2 1.3 *Strapping* - Strapping (when required) shall be in accordance with applicable box specification

5.2 2.2 *Level B*

5 2.2.2 1 *Exterior container* - The number of units specified in table VI shall be overpacked in an exterior container constructed in accordance with Federal Specification PPP-B-621, table II

5.2 2.3 *Level C* - The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order

5 2.3 *Marking* - Unit and intermediate containers shall be marked in accordance with section 5 of Military Standard MIL-STD-129

5 2.3 1 Shipping container(s) shall be marked in accordance with Military Standard MIL STD-129 Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities

6 NOTES

6 1 Ordering data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following

(a) Title, number, and date of this specification
(b) Style and size required (see 1 2)
(c) Selection of applicable levels of packaging and packing for military procurement (see 5 1 and 5 2)

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification If, for any particular purpose, samples with bids are

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necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects

6.3 Federal Specifications do not include all types, classes, grades, sizes, etc of the commodities indicated by the title of the specification, or which are commercially available, but are intended to cover the types, etc , which are suitable for Federal Government requirements

6.4 Transportation description - Transportation description applicable to this item is

Sets, Parenteral Administration, Expendable
Carload minimum weight 24,000 pounds subject to Rule 34, Uniform Freight Classification

Sets, Expendable Administration for Parenteral Products
Truckload minimum weight 10,000 pounds, subject to Rule 34, National Motor Freight Classification

Notice - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof)

SPECIFICATION

ORGANIZATION (Of submitter) | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

5 SUBMITTED BY (Printed or typed name and activity) | DATE

FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO 15419

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DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

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PLATE NO 15419 (BACK)

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